

Scenario 14

A researcher needs to simulate a network with a specific bandwidth limitation. They want the link between host h1 and switch s1 to be limited to 10 Mbps.

Question: How would they specify this link parameter when starting Mininet from the command line?

Answer:

To specify a global link bandwidth limit when starting Mininet from the Linux command line, the researcher would use the `--link` flag set to `tc` (Traffic Control) alongside the `bw` parameter:

```
sudo mn --link tc,bw=10
```

Explanation & Alternatives:

- **Global Command Line Parameter:** Running `sudo mn --link tc,bw=10` configures Mininet to use Linux Traffic Control (TC) and sets the bandwidth of all default links (including the link between h1 and s1) to 10 Mbps.
- **Custom Python Topology (For Specific Single Link):** If the researcher wants *only* the link between h1 and s1 to be 10 Mbps while keeping other links at default speeds, they must use a Python script with `TCLink`:

```
from mininet.link import TCLink
net.addLink(h1, s1, cls=TCLink, bw=10)
```